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Pressure and temperature measurement in food process control

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10.1 Introduction

Chapters 10 and 11 are concerned with the measurement of process variables that are important to the manufacturing of food. Although not giving a direct indication of product quality, accurate measurements contribute to better process control. This in turn affects the reproducibility of the manufacturing process, its economy, and the durability of the final product.

Here we shall be concerned primarily with the variables pressure, temperature, flow and level, but other measurements may also be required for full control. Thus trace moisture sensors are essential for monitoring the water content of blanket gases or dry air, and humidity control is required in the conditioning of products such as stored fruit, meat or other raw materials. In-line pH measurements are often encountered in the manufacture of milk products and conductivity measurements are required in the production of pure water. Occasionally, Redox and turbidity measurements may also be employed. As no further details will be given in this chapter of the methods used to measure these variables, the interested reader is referred to other literature on the subject [1].

This chapter covers the fields of temperature and pressure measurement, two process variables that account for over 80% of the measuring points in the processing industries. Over the years a great many instruments have been developed to measure them, but in this book only those that might be met in a food production facility will be considered.

Hygiene is an ever-present factor in food production, and a number of guidelines have been developed to assure the quality of the end product. Those especially applying to process instrumentation are discussed in Section 10.4. Here, instrument design is considered from the aspects of process connections, materials and cleaning-in-place.

Finally, Section 10.5 looks at a few typical applications for temperature and pressure instruments. These include descriptions both of food processing chains and ancillary processes, since a food manufacturing plant not only produces the end product but must also make provision, e.g. for compressed air, refrigeration, steam heating as well as pure and waste water.

10.2 Pressure measurement

In food processing, pressure measurements are required in piping, across filters, as well as in closed tanks. For piping in particular, the instruments must be designed to withstand overload pressures far beyond their normal operating range. As liquids or solids are pumped along the pipe, the pumping action causes regular surges in pressure. When valves are opened and closed, the pressure may increase to the normal operating pressure of the pump. If a valve closes abruptly, as is often the case, high pressure peaks are generated – a particular problem in the low pressure region. Finally, pressure instruments must be immune to vibration and be able to withstand the temperatures and stresses induced by internal and external cleaning.

Pressure measurement devices are categorised according to the type of pressure they measure: absolute, gauge and differential pressure.

- Absolute pressure devices measure the actual pressure acting on the sensor. For atmospheric pressure, therefore, they display approximately 1000 millibar.
- Gauge pressure devices measure the pressure relative to atmospheric pressure. In this case, atmospheric pressure would produce a reading of 0 millibar – lower pressures are registered as negative values.
- Differential pressure devices measure the pressure difference between two tapping points, e.g. in a pipeline or a tank. They are usually used to measure level and flow.

10.2.1 Measurement methods

As far as the measuring principle is concerned, practically all pressure gauges fall under the category force-type pressure devices. The way in which the force resulting from the acting pressure is sensed leads to a further subdivision into:

- Manometer-type measurement devices
- Mechanical-type measurement devices
- Electrical pressure transducers.

Of these methods, electrical measurements are most suited to automation, although the other types are still encountered in many plants.

10.2.2 Manometer-type instruments

A manometer, see Fig. 10.1, comprises a U-tube filled with a so-called manometric fluid (water, oil or mercury), one limb being connected to acting pressure and the other being open, closed or otherwise connected to an indicating, registering or recording mechanism. The pressure to be measured acts upon the liquid in one limb of the manometer, pushing it into the other limb. The liquid continues to be displaced until the forces acting in both limbs are equal. For a gas, the difference in the liquid levels in the two limbs of the manometer gives a reading of acting pressure.

So-called well or J-tubes indicate pressure directly. Here the diameter of the pressure limb is much greater than the measuring limb, so that a minimal change in level causes a large change on the measuring side. Where small pressure differences are to be measured, the tube may be inclined to give more accurate readings. Other designs use a float and mechanical assembly which operate a pointer when the manometric fluid level changes.

In addition to classical manometers, there are numerous more sophisticated variations, e.g. balance hollow ring, bell chamber or bellows. In these devices, the manometer fluid

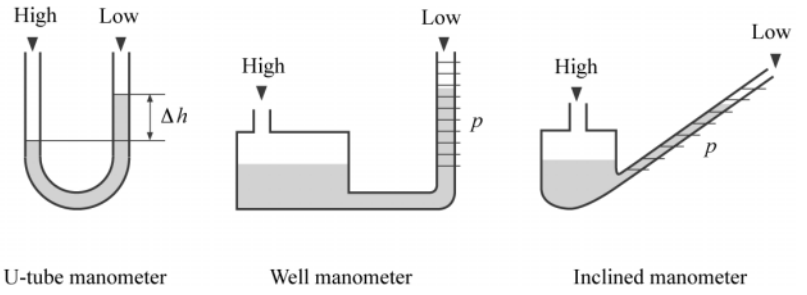


Fig. 10.1 Manometer-type pressure measurement devices.

is used to separate the pressure limb from the measuring limb, on which acts a reference pressure. The difference in pressure is transmitted by a mechanical assembly to a dial gauge.

In view of their simplicity, manometers are still a common sight in many plants, where they provide an optical control of the pressure. They are, however, losing ground to the other measurement methods. Classical manometers are also sensitive to temperature changes and must therefore be used within a fairly strict temperature range. Considering that the manometric fluid is in direct contact with the process, they are not suitable for sterile applications.

10.2.3 Mechanical-type measurement devices

Mechanical-type measuring devices translate the pressure acting upon them into a mechanical displacement. There are various designs of measuring element, of which diaphragms and (Bourdon) tubes are more likely to be met with in the food industry, see Fig. 10.2.

- A diaphragm element is a thin, flexible metal plate that separates the chamber connected to the pressure being measured from the atmosphere. It may be flat, corrugated or in the form of an introverted capsule. When both sides of the diaphragm are exposed to the same pressure, the diaphragm is in its equilibrium position. If a negative pressure is applied (less than atmospheric) the diaphragm is pulled in, if a positive pressure is applied, it is pushed out. The maximum displacement occurs at its centre.

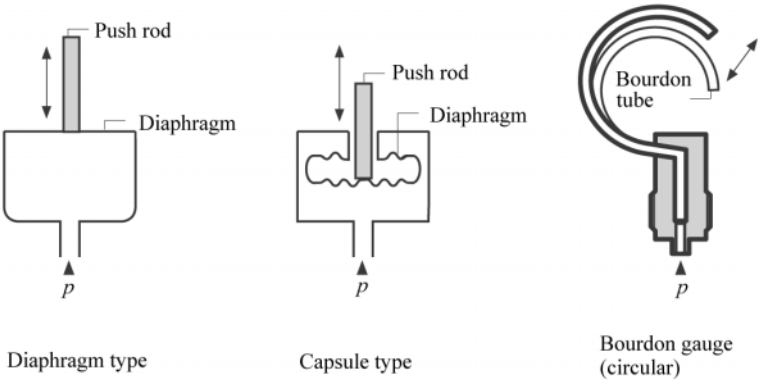


Fig. 10.2 Mechanical-type pressure measurement devices.

- A Bourdon tube element comprises a thin flattened metal tube, sealed at one end, which is connected to the pressure being measured. The tube is bent into a circlar or helical form, or may simply be twisted. When positive pressure is applied, the tube will tend to straighten out, producing a linear or radial deflection that is proportional to the acting pressure.

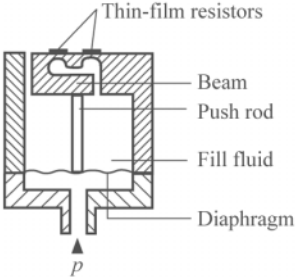
By the use of a suitable mechanical assembly this displacement or deflection can be used to give an indication of pressure, e.g. on a dial. If the dial is fitted with electrical contacts, pressure limits can monitored. The pointer and dial display is easy to read and gives a quick indication of pressure. In automated plant, however, most serve only as an optical back-ups to electrical pressure transmitters.

10.2.4 Electrical pressure transducers

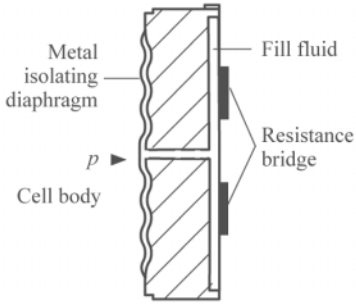
The majority of electrical pressure transducers use a flexible diaphragm as the pressure transmitting element. Unlike mechanical-type devices, the diaphragm is not a part of a pressure chamber connected to the process, but forms the front isolating element of a sensing chamber. Thus, by using the appropriate process connection, it is possible to produce flush mounted pressure devices that have no cavities and that are easy to clean.

Figure 10.3 shows six typical measuring cell designs for absolute/gauge and differential pressure measurement.

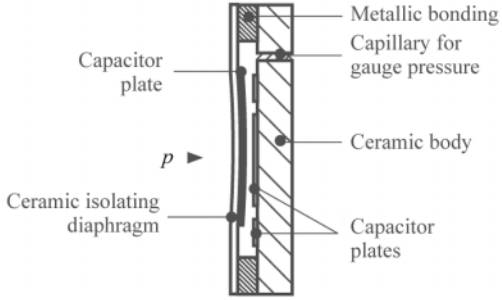
- Type A is a resistive gauge/absolute pressure cell. The pressure acting on the isolating metal diaphragm is transmitted by a push rod onto the flexible beam, which deflects. Strain gauges located on the beam generate an electrical signal proportional to the acting pressure. The absolute cell is evacuated, the gauge cell is filled with fill fluid.
- Type B is a resistive gauge/absolute pressure cell. The pressure acting on the isolating metal diaphragm is transmitted by a fill fluid to a second, flat, sensing diaphragm. The sensing diaphragm is also deflected, whereby the degree of deflection is greatest at its centre. Strain gauges or piezoelements on its surface produce an electrical signal proportional to the acting pressure. If the diaphragm is overloaded it sets back on the sensor body, avoiding any mechanical damage. The absolute cell is evacuated, the gauge cell is filled with fill fluid.
- Type C is a capacitive gauge/absolute pressure cell. Two annular electrodes are to be found on the body, a third on the inner face of the diaphragm. When the diaphragm deflects, the capacitance between the inner body and diaphragm electrode changes in proportion to the acting pressure. Since there is virtually no movement towards the outer body electrode, this outputs a constant reference signal. If the diaphragm is overloaded it sets back on the sensor body, avoiding any mechanical damage. The cell is completely dry. The absolute pressure sensor is sealed and evacuated.
- Type D is a twin-chamber capacitive differential pressure cell. It comprises two corrugated isolating diaphragms on the low and high pressure sides. To prevent damage on overload, these lie back on a similarly formed substrate. The space between the substrate and isolating diaphragm as well as the entire sensing capsule is filled with fluid. Capillaries lead from the front to the centre of the cell, where a sensing diaphragm separates the two sides. Metal electrodes are deposited on the diaphragm and capsule walls, forming the capacitor which measures the pressure when the sensing diaphragm deflects.



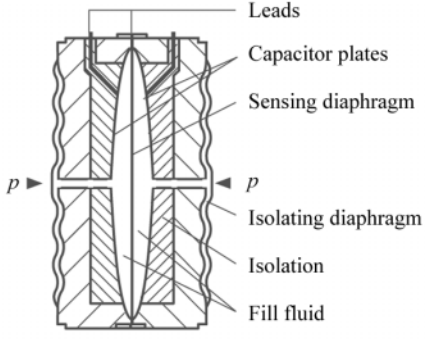
Type A: Resistance measuring cell with push rod and beam



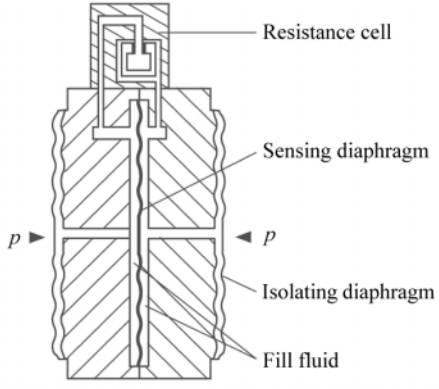
Type B: Resistance measuring cell



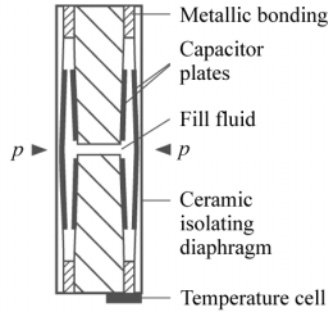
Type C: Dry capacitance measuring cell



Type D: Capacitance Δp measuring cell (twin chamber)



Type E: Resistance Δp measuring cell



Type F: Capacitance Δp measuring cell (ceramic single chamber)

Fig. 10.3 Measurement cells for electrical pressure transmitters.

Inductive cells have a similar design, whereby the metal sensing diaphragm has two ferrite discs fixed to each side. These together with two fixed magnetic coils on each side of the sensing capsule body form a differential transformer. When the sensing diaphragm deflects, there is a change in inductivity that is proportional to the acting pressure.

- Type E is a twin-chamber resistive cell for differential pressure. The construction of the cell is essentially the same as the twin-chamber capacitance cell, with the exception that additional capillaries lead to an external sensing capsule. Any deflection at the isolating diaphragms is transmitted by the fill fluid to the external sensing diaphragm. Strain gauges or piezoelements then produce a pressure proportional electrical signal.
- Type F is a single-chamber capacitive cell for differential pressure cell. The two sides of the cell are connected by a capillary and a fill fluid transfers the pressure from one side to another. The cell body and diaphragms are ceramic, each side of the cell having its own set of capacitance electrodes, which deliver an electrical signal which is a function of the pressure. Since the two sides are connected, it is possible to build a self-monitoring cell. As the temperature increases, the volume of the fill fluid increases and the capacitance of the sensor changes. Since the expansion coefficient of the fill fluid is known, this gives an indirect measurement of temperature. By comparing this temperature with an independent measurement of the sensor temperature, it is possible to detect whether the sensor is faulty.

There are many other transducers on the market, e.g. resonance-wire or vibrating-beam, but these are often associated with a particular vendor and are beyond the scope of this book. There is also a further subdivision within the resistive type cells which relates to the sensing element.

- So-called metallic sensors use strain gauges of various types. Foil strain gauges are made from resistive alloys such as Konstantan that have been rolled into thin foils a few μm thick, coated with an isolating organic substance and stuck to a metal diaphragm. Thin-film strain gauges are produced by vapour deposition, spluttering or chemical vapour deposition techniques.
- Piezoresistive sensors are based on semiconductors that have pressure-dependent electrical properties. There are two basic types of element. *Polysilicon sensors* are manufactured in the same way as metallic thin-film strain gauges and are doped to produce the desired electrical properties. *Monosilicon sensors* are made of pure silicon and are anisotropic, i.e. their resistance is direction-dependent, allowing them to produce highly sensitive sensors.
- Thick-film strain gauges are produced by silk-screen printing techniques. The individual layers required to produce the desired sensor geometry are built up on a ceramic substrate using several different masks.

10.2.5 Summary

Table 10.1 summarises the typical operating conditions of the various measurement methods. The accuracy of measurement is not quoted since it depends on the individual transmitter: in general it comprises the measured error, the hysteresis and reproducibility measured at a reference temperature, usually 20ºC. Often all three are quoted as an 'overall' accuracy. Also to be considered is the temperature effect, which indicates by how much the accuracy changes for a given temperature rise. This can be quite

Table 10.1 Typical operating conditions of various process pressure devices. The values are guidelines only, for specific values see the manufacturer's data sheets. For electrical transmitters, the measuring range can often be 'turned down' by 40:1 or more

Type	Temperature range	Min./max. pressure range	Differential pressure range	Remarks
Manometer	−10ºC to +60ºC	0 mbar to 1 mbar 0 bar to 20 bar	–	Range depends on manometric fluid. Inaccurate if not operated within the rated temperature range.
Diaphragms	Up to 100ºC	0 mbar to 10 mbar 0 bar to 25 bar	–	Significant temperature effect Up to 5 × overload (max 40 bar)
Capsule	−25ºC to +60ºC	0 mbar to 2.5 mbar 0 mbar to 600 mbar	–	Significant temperature effect
Bourdon gauges	−20ºC to +60ºC	0 mbar to 600 mbar 0 bar to 4000 bar	–	Significant temperature effect
Electrical resistive	−30ºC to +200ºC	0 mbar to 1 bar 0 bar to 400 bar	Up to 3 bar for static pressures up to 140 bar	Actual pressure and temperature range depend on cell design. Usually resistant to vibration and overload.
Electrical capacitive	−30 ºC to +200ºC	0 mbar to 400 mbar 0 bar to 40 bar (ceramic) 0 bar to 600 bar (metal)	Up to 3 bar for static pressures up to 420 bar (ceramic) Up to 500 bar for static pressures up to 500 bar (metal)	Actual pressure and temperature range depend upon cell designs. Resistant to vibration and overload. Ceramic cells are more corrosion resistant and self-monitoring.
Electrical inductive	−20ºC to +120ºC	0 mbar to 100 mbar 0 bar to 160 bar	Up to 160 bar for static pressures up to 210 bar	Actual pressure and temperature range depend upon cell designs. Resistant to vibration and overload.

considerable for certain fill fluids or if the sensor is operating a with diaphragm seal (an extension which allows flush mounting) or a remote seal and capillaries. The overload pressure indicates the ability to withstand pressure peaks without mechanical damage to the sensor; a re-calibration may be necessary, however. More information on the testing of pressure and other process transmitters can be obtained from IEC Standard 61298, Parts 1 to 4 and DIN 16086 [2, 3].

10.3 Temperature measurement

Temperature is by far the most frequently measured variable in process engineering. In food production, the monitoring and control of temperature is an important factor in assuring the quality of the end product. Two examples illustrate why this is so:

- Storage silos are often monitored, because any unaccountable rise in temperature may be the first indication that the raw material in store is deteriorating.
- The maintenance of a wrong temperature during the manufacture of dairy and other products seriously affects their shelf-life.

Although there are many applications for temperature measurement in food processing, the process conditions to be encountered are not particularly hostile. Thus, the temperatures to be met range from about −50ºC in cold storage to +150ºC in sterilisation-in-place applications. Only in steam generation will higher temperatures be found. Accuracy is very important, and of course, hygiene is essential – see Section 10.4.

10.3.1 Measurement methods

Temperature measuring devices may be divided into three different categories according to the method of measurement they employ: force temperature devices, electrical temperature devices and radiation pyrometer temperature devices. Standard pyrometers are usually encountered in high temperature applications, e.g. in furnaces, so will not be considered further in this chapter. More on infra-red thermometry can be found in Chapter 7.

10.3.2 Force temperature devices

Force temperature devices (FTDs) can be further divided into bimetallic FTDs and filled thermal system FTDs, see Fig. 10.4. They make use of the fact that the length or volume of a given mass of matter changes as its temperature increases. The change in length or volume with unit temperature is dependent upon the material and is characterised by a so-called coefficient of expansion.

Bimetallic strip thermometers

A bimetallic strip comprises two strips of metal with different coefficients of expansion that are bonded together, e.g. by riveting. When the temperature rises, each metal expands by a different amount causing the whole strip to bend. The amount of deflection is an indication of temperature. By a suitable mechanical assembly, this deflection can be used to give a continuous indication of temperature at a graduated display or to make contact with a switching element.

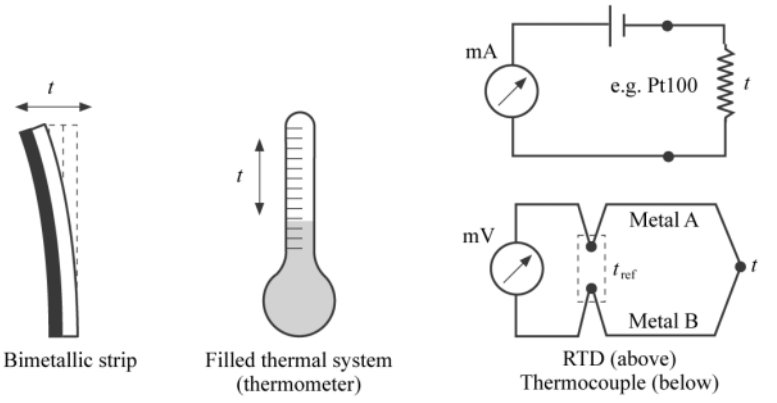


Fig. 10.4 Temperature measurement device designs.

Bimetallic strip gauges are inexpensive and good designs provide reasonable accuracy (±2%−±3% of full scale (FS)). They are built for specific temperature ranges. Continuously reading bimetallic thermometers are often used as optical back-ups to resistance thermometers and in applications where temperature is not used as a control variable. Bimetallic switches are often found in two-point temperature controls.

Filled thermal systems

Filled thermal systems make use of the thermal expansion of liquids to provide a direct indication of temperature. In the most common of all, the glass bulb thermometer, the bulb acts as a reservoir for the liquid and is connected to a graduated capillary tube. As the temperature rises, the liquid within the bulb expands into the capillary; the height of the liquid column indicates the current temperature. The liquids used are, e.g. coloured alcohol or mercury. Thermometers are designed to measure over a specific range, usually between 0ºC to 100ºC (or the Fahrenheit equivalent). If the flow of liquid back to the bulb is restricted, e.g. by a kink in the capillary, they can also be designed to give more accurate measurements over smaller ranges.

Should a glass-bulb thermometer break, glass splinters and alcohol or mercury would enter the process and contaminate the product. For this reason, thermometers are mounted in protective metal sheaths. Here it is essential that there is good thermal contact between the bulb and the sheath, otherwise the temperature indication is inaccurate and the response slow. Alternatively metal bulb thermometers might be used. The accuracy of a glass-bulb thermometer depends on the graduated scale and observer, but is generally within ±1ºC. It is essentially built for the human observer and does not lend itself easily to automation. For this reason its use today is limited to optical back-up of electrical systems or applications in which the temperature is not a control variable.

Liquid spring thermometer

A variation of the glass-bulb thermometer is the liquid spring thermometer. Here the metallic bulb and capillary are completely filled with liquid. As the temperature of the system increases, so does its pressure. The pressure changes are transmitted via a metal diaphragm to an elastically deformable spring with pointer, thus giving a direct reading of temperature. The latest versions of this type of thermometer are now coupled directly to a pressure sensor, which allows them to be integrated into process control systems.

10.3.3 Electrical temperature devices

Due to the ease with which they can be integrated into control systems, electrical devices are the preferred method of temperature measurement in process control. They use the dependency of electrical properties of particular materials on temperature to provide a measurement, see Fig. 10.4. There are four different types: semiconductors, resistance temperature detectors (RTDs), thermocouples and silicon resistors, whereby only RTDs and thermocouples are of real interest in the production of food.

Resistance temperature detectors

Resistance temperature detectors (RTDs) or resistance thermometers are the most common type of temperature sensor to be found in process engineering. They comprise a thin-film or wire resistor with a standard resistance of 100Ω, 500Ω or 1000Ω. The resistor material may be platinum or nickel, the standardised designations being Pt100, Pt500, Pt1000 or Ni100. The sensors are very stable, have a wide operating range from −200ºC to +850ºC depending upon type and exhibit a well defined relationship between resistance and temperature. As a consequence they are interchangeable over a wide temperature range.

RTDs may have a relatively slow time response and offer only a small change in resistance per unit increase in temperature. To limit or avoid errors in the determination of such small changes in resistance for measuring points where the sensing element and the evaluating electronics are separated by some distance, three- or even four-wire measurement technology must be used. RTDs are also sensitive to vibration and shock, so that care must be in their design.

Thermocouples

If two wires of different metals are joined together at their ends and the two junctions are at different temperatures, a potential difference is created that causes a current to flow around the loop. The difference in potential arises because the magnitude of the contact potential at each junction is dependent upon the temperature. If one of the junctions is kept at a constant temperature, the magnitude of the current that flows is a measure of the temperature at the other.

Although thermocouples have a wide operating temperature range, in Europe they are usually only used in preference to RTDs for temperatures above 600ºC. In America, however, they are being increasingly used for lower temperature ranges. They can measure up to +1800ºC depending upon the type and conductor diameter, and usually have a faster response than RTDs but a worse accuracy. They are simple, rugged, inexpensive and require no external power supply. On the other hand, the signal is non-linear, they exhibit low sensitivity and have relatively low stability. A reference junction is required and they must be compensated. Moreover, metallurgical changes and ageing sometimes causes a loss of performance. A more serious problem is that the low voltage output is susceptible to electromagnetic interference.

10.3.4 Design and assembly of RTDs

Since RTDs are the temperature sensors most likely to be found in the food industry, their design, assembly and use will be examined in more detail. The design has features in common with other electrical temperature devices, since most comprise the sensing element in its protective sheath, a thermowell for introducing the sensor to the process and a head or remote transmitter for evaluating and processing the incoming signal as

well as providing power to the sensor circuit. The use of a thermowell is also common practice for force temperature devices.

Sensing element
There are various designs of Pt100 sensing elements.

- The traditional sensing element is the ceramic Pt 100 element which comprises a spiral of extremely fine platinum embedded in ceramic powder (frit) within capillaries of extremely pure alumina. Two platinum wires provide the contacts to the connecting wires.
- The glass Pt 100 element comprises an extremely thin platinum tape wound around a glass mandrel. The element is protected by a glass sleeve which is fused to it. Platinum wires provide the contacts to the connecting wires. Due to the fusion of the sheath and element, this design has the advantage of a high vibration resistance.
- For the thin film Pt 100 element, an extremely thin layer of platinum is evaporated onto a 0.5 mm thick ceramic substrate. By using a laser or etching technique, platinum is removed selectively from the surface in order to create a meander with a nominal resistance of 100Ω at 0ºC. The platinum coated connecting wires and thin film are protected by a fused glass coating.

Table 10.2 lists the nominal operating temperature ranges for the three types of Pt 100 sensing element.

The resistance measured by the RTD sensing element is converted to a temperature by using the tables in European Standard EN 60 751 (= IEC 60 751) [4]. These values have been determined on the basis of the International Temperature Scale ITS90 [5], and can be calculated by using a polynomial equation, the coefficients of which are exactly known for each sensor element type, i.e. for Pt100, Pt500, Pt1000 and Ni100. The standard further defines two accuracy (tolerance) classes for Pt 100 sensing elements:

- Class A: accuracy in ºC = 0.15 + 0.002 (t)
- Class B: accuracy in ºC = 0.30 + 0.005 (t)

where (t) is the unsigned numerical value of the temperature in ºC. This classification is often not adequate for industrial users and it is common to meet fractions or multiples of the classes. For example, Class 1/3 B would correspond to an accuracy in ºC of 0.10 + 0.005 (t) and 2 × B to 0.60 + 0.005 (t). For fractional B classes, the temperature dependent part of the tolerance (0.005 (t)) sometimes has a lower value, thus assuring greater accuracy.

Type of connection
RTD sensing elements are manufactured by a few specialist firms around the world. The top five deliver products of comparable high quality to most of the major instrument manufacturers. It is only in the design beyond the element, that differences are to be

Table 10.2 Nominal operating temperatures of Pt100 sensing elements

Type	Dimensions	Design	Isolation	Temperature range
Pt100	2.5 mm dia × 35 mm	Wound	Ceramic	−200ºC to 850ºC
Pt100	2.5 mm dia × 35 mm	Wound	Glass	−200ºC to 600ºC
Pt100	10 mm × 2 mm × 0.5 mm	Thin film	Glass/ceramic	−50ºC to 400ºC

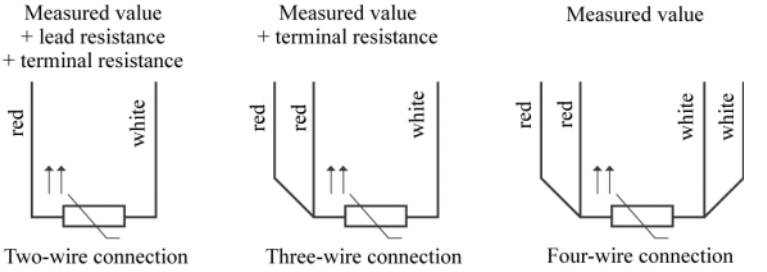


Fig. 10.5 Types of connection for RTDs.

found. The two connecting wires of the sensing element carry the signal to the evaluating circuit which in its simplest form might comprise a constant current source and a means of measuring the resistance. For process instrumentation this circuitry is usually found in a head or remote transmitter. Since the evaluating circuit and sensing element are always removed from one another, the gap has to be bridged by connecting wires. If the whole sensing assembly is to be removable, e.g. for replacement in the event of a failure of one component, then it is usual that these wires end in terminals. Connecting leads are then required to carry the signal to the evaluating circuit. The connecting wires, leads and terminals may all be sources of error in the final measurement. For this reason, two-, three- and four-wire circuits are to be found, see Fig. 10.5.

For a two-wire circuit, the resistance of the connecting wires, leads and terminals must be added to the resistance of the sensing element. The longer the leads, the higher the resistance and the larger the resulting error. For this reason two-wire circuits should be used only when the transmitter is located close to the sensing element. This is particularly critical for Pt100 and Ni100 RTDs.

In a three-wire circuit, an additional wire is connected to one side of the sensing element. This is connected to the transmitter and acts as a reference for the lead resistance. Provided the leads are of identical material and length, this reference value can be subtracted from the detected resistance to provide a measurement that is influenced only by the terminal resistance. The remaining error is roughly 0.1% of the measured value, which provides sufficient accuracy for most industrial applications.

The four-wire circuit is used for very accurate measurements. Two connecting wires on each side of the sensing element are connected to the transmitter. A constant current is supplied to the element via one pair of wires, whilst the other pair are used to measure the resulting voltage. By using this type of voltage compensation, unbalanced wires and leads, and their associated resistances, have no effect on the measurement.

Intrinsic warming (self-heating)

In order to measure temperature, an RTD must be supplied with a constant current. This current also causes the device to heat up. The degree of heating is dependent upon the design, and is particularly critical for thin-film elements. It is extremely difficult to determine the extent of this so-called intrinsic warming or self-heating, however, for a Pt100 sensor operating at 10 mA it is typically 0.02 K to 0.4 K in water and 0.9 K in air. For this reason it is usual to operate the RTD at a constant current of 1 mA, making the intrinsic warming effect negligible in comparison to the standard accuracy.

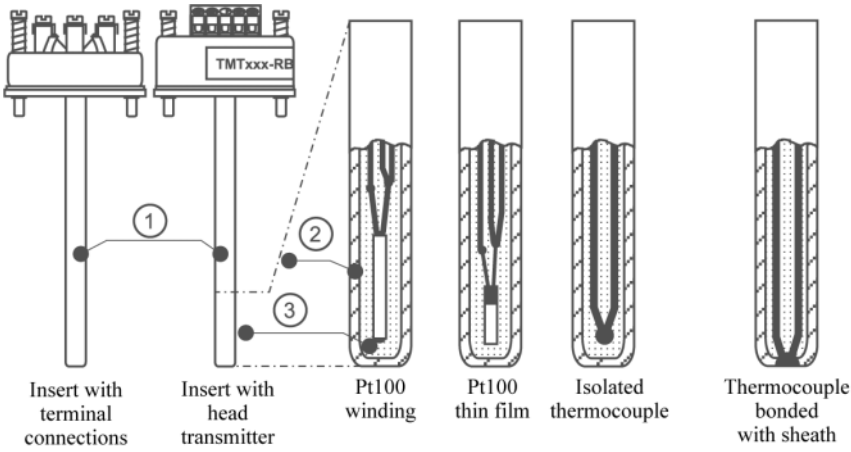


Fig. 10.6 Contruction of sensor insert for RTDs and thermocouples; 1 leads with ceramic isolation, 2 metal or ceramic sheath, 3 alumina powder.

Sensor insert

Before the sensing element can be used in an industrial application, it must be protected against shorting and vibration. To this end, it is built into an exchangeable sensor insert (or inset). The insert comprises a metal sheath, usually AISI 316 stainless steel, containing the sensing element and connection wires, see Fig. 10.6. To prevent them from shorting, the connection wires are threaded through thin ceramic tubes. The whole assembly is then packed in alumina powder to protect it against vibration.

A more advanced insert manufacturing technique uses magnesium oxide cables, i.e. extruded metal tubes that contain the connecting wires isolated by compressed MgO powder. This method has the advantage of providing excellent electrical isolation and provides protection against environmental humidity. Furthermore the insert is flexible, vibration-proof and can be operated at high temperatures. The top of the insert may be of two different types. Either the connecting wires are led out via ceramic sockets to a set of terminals or they are connected directly to a head transmitter which contains the evaluating circuits. The number of terminals depends upon the type of connecting circuit used.

Thermowells

For most applications, the sensor insert is not suitable for direct contact with the process. This is the case for food manufacture, where the external surface might offer conditions for bacterial contamination. The insert is therefore operated within a thermowell, which is often an integral part of the sensor assembly, see Fig. 10.7. For food processing, this comprises a highly polished, closed, stainless steel tube with a process connection or welding neck. In order to ensure the best possible transfer of heat, the sensor insert is usually in contact with the well bottom or set into a thermoconductive paste and often tapers towards the end to provide better heat transfer.

Head or remote transmitter

The transmitter is the brains of the sensor and converts the signal from the sensing element into a standardised output. It also powers the sensor where necessary. Modern transmitters are equipped with a microcontroller which provides a direct conversion of resistance (or voltage) to temperature for output at a local display or to a fieldbus system,

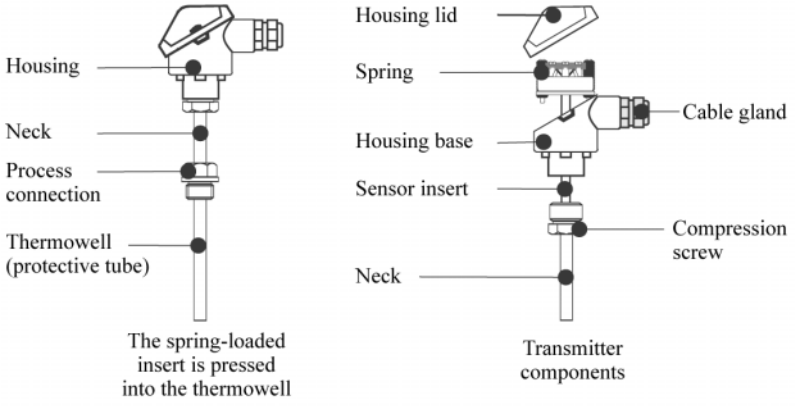


Fig. 10.7 Construction of a temperature sensor, here shown with sensor insert with terminal connections.

as well as compensation functions, e.g. for non-linearity of the sensor signal. Alternatively, a simple temperature-proportional 4...20 mA or 0...10 V signal with freely assignable temperature range end values may be provided. It is usual that either RTDs, thermocouples or voltage signals can be connected to the same transmitter.

If the transmitter is to be found in the sensor assembly, it is known as a head transmitter. If on the other hand, the transmitter is not an integral part of the sensor, i.e. it is located in a nearby control cabinet or in the control room, it is known as a remote transmitter. Since the sensing element of all electrical temperature devices outputs an electrical signal, it is also possible to make a direct connection to an output device such as a chart or data recorder which has the facilities to handle temperature signals.

Housing

The housing of a temperature device must protect the sensor insert, the terminals and, where appropriate, the head transmitter from the ingress of water and dust. It must also provide good access to the terminals and sufficient space for the wiring. The size therefore varies according to type of transmitter. The connecting leads to the sensor are lead through a cable gland into the inside of the housing. All housings have either an IP or NEMA rating which indicate the degree of protection they allow when the lid is firmly clamped or screwed into position and the cable gland screwed tight, see Section 10.4. For food applications, housings are often made of stainless steel, although polypropylene (FDA approved) is often encountered in North America. Standardised housings are to be found in DIN 43 729, whereby Type B is the one most often used [6].

10.3.5 Installation conditions

Even the most accurate temperature sensor will deliver incorrect measurements if it is not properly installed. As mentioned previously, for food applications electrical temperature devices will usually be installed in thermowells. These in turn will be built into the pipes and tanks where the medium to be measured is flowing or stored. Since the conditions to be met at the measuring points are different, e.g. gases, liquids or solids, flowing or stationary, various designs of thermowell are available for different measurement tasks.

For measurements in a flowing medium, the thermowell will have been designed for a particular range of flowrates. It is beyond the scope of this book to discuss thermowell design further, but more information is available from vendor's catalogues.

Heat transfer

Electrical temperature devices must always be in good thermal contact with the medium to be measured so that any change in temperature is recorded as quickly as possible. Since the heat transfer coefficients of solids, liquids and gases differ considerably, each application requires the thermowell to installed in a different way. Adequate heat transfer is assured by ensuring that a certain length of the thermowell, the so-called immersion depth, is exposed to the medium.

If the immersion depth is too shallow, heat may be dissipated to the pipe or tank, resulting in an incorrect measurement. This phenomenon is enhanced when the sensing element is placed close to the process connection; the metallic mass radiates heat to the external environment. To avoid measurement errors, therefore, the minimum immersion depth should be at least 15 times the diameter of the thermowell tip, whereby, if possible 20 times the diameter should be used. In all cases, the technical data sheets and standards governing thermowells should be consulted.

Response time

A second important factor in temperature measurement is the speed with which a change in temperature is registered by the sensing element. Since this also depends on the rate of heat transfer and the thickness of the thermowell tip, tests are made under standard conditions. The characterising variable in EN 60751 is the response time [4], also called the settling time in the instrument test standards series IEC 60770 and IEC 61298 [7, 2]. The response time is the time the sensor output requires to reach a certain percentage of its final steady-state value after a step rise in the input temperature. The times are designated $\tau_{0.5}$ for 50% steady-state value and $\tau_{0.9}$ for 90%.

The response time is determined under two sets of conditions:

- In air flowing at a velocity of 3 m/s, immersion depth 15 $\times$ the diameter of the thermowell tip plus the sensor element length, start temperature between 10ºC and 30ºC, temperature jump between 10ºC and 20ºC.
- In water flowing at a velocity of 0.4 m/s, immersion depth 5 $\times$ the diameter of the thermowell tip plus the sensor element length, start temperature between 5ºC and 30ºC, temperature jump not greater than 10ºC.

Table 10.3 lists reference values for typical thermowell thicknesses. The response time decreases with increasing flowrate and vice versa.

Mounting position

Table 10.3 shows clearly the relationship between response time, thermowell diameter and immersion depth: the thinner the thermowell, the smaller the immersion depth and the quicker the response. On the other hand, the thermowell acts as a resistance to air or liquid flow and is subject to mechanical stresses and vibration. In some cases, it might also be subject to abrasion or chemical attack. These factors have to be weighed against each other when selecting a suitable thermowell. The choice of mounting position may also influence this decision by producing more favourable conditions.

When temperature sensors are mounted in pipes, then a position or orientation should be chosen that reduces the surface area facing the direction of flow. This reduces the

Table 10.3 Response times under standard conditions for various thermowell thicknesses

Diameter of tip	In water at 0.4 m/s		In air at 3 m/s	
	$\tau_{0.5}$	$\tau_{0.9}$	$\tau_{0.5}$	$\tau_{0.9}$
9 mm	30 s	80 s	133 s	390 s
10 mm	34 s	103 s	144 s	398 s
11 mm	38 s	125 s	154 s	405 s
12 mm	44 s	143 s	180 s	430 s
13 mm	50 s	160 s	205 s	455 s

mechanical load on the thermowell. Since the sensor produces turbulence in the downstream flow, care should be taken that it is not positioned too close to any other instrumentation in the pipe.

- The ideal position is in a pipe elbow. This offers the smallest cross-section and least mechanical load.
- Where the pipe is wide enough, the sensor can be mounted vertically.
- The sensor can be angled towards the oncoming flow if the pipe diameter does not allow vertical mounting.
- For very small pipes, the sensor can be mounted on a nozzle which is wide enough to allow fluid to flow around the thermowell.

For tanks, the situation is somewhat better, in that there is usually space for vertical or horizontal mounting. The temperature sensor must be positioned such that it is covered with medium at all times when the signal is required. Care should be taken not to position it too close to moving parts such as agitators, pumps or valves, however, since the forces generated by the movement of fluid may exceed the mechanical strength of the thermowell or lead to increased abrasion or corrosion.

10.4 General instrument design

The production of food is one of the most carefully controlled areas of industrial activity. In addition to being strictly controlled by government agencies, the industry itself has set up its own watchdogs to monitor advances in production engineering and make recommendations regarding safe, modern practice. The aim, of course, is to protect the consumer from exposure to unsafe products, resulting from the presence of e.g. foreign bodies, chemical contaminants or bacteria. The use of instrumentation manufactured according to these guidelines is a step towards safe food production.

10.4.1 Design factors

When choosing instrumentation for food production, it is always wise to bear in mind that it is not the normal operation of a device that gives problems, but rather the unexpected event. Thus, the risk of chemical contamination can be eliminated by using a suitable material for the wetted parts of the device. The risk of bacterial contamination can be reduced by regular cleaning and the use of suitably designed process connections. The introduction of foreign bodies, however, is only partially covered by the adoption of a high degree of protection. The case where equipment in direct contact with the product fails,

producing debris or releasing contaminants, must also be considered. Here it is essential that the user is warned and/or that the released products are not dangerous to health.

10.4.2 Wetted parts

The wetted parts of a device are those parts which are in contact with the medium being measured. For temperature measurement this might be the thermowell, for pressure measurement the isolating diaphragm and for a contacting level measurement the sensing element itself. Even so-called non-contact devices must be considered to have wetted parts when they intrude into the pipeline or tank. Here it is not so much the contact with the medium, but crevices and their exposure to high temperatures and vapours which has to be considered.

The positioning of the measurement device must also be examined. Flowing gases, liquids or solids may cause abrasion or generate high mechanical forces, which combined with high temperature or vibration enhance electrochemical attack or mechanical fatigue. Moreover, the wetted parts must be able to withstand the forces and temperatures generated during cleaning or sterilisation-in-place procedures.

Materials

In addition to the normal mechanical design factors, the toxicological and bacteriological compatibility of the materials used for wetted parts must also be taken into consideration. As far as the toxicological properties are concerned, a material approved by the USA Federal Drug Administration (FDA approval) or equivalent regulating body must be used.

The bacteriological factor is a different matter. Although regular cleaning and gap-free design reduce the broad risk of infection, the proper design and finishing of the wetted parts is just as important. This basically means flowing contours and clean welding, no nooks and crannies, and no obstructions that might cause the product to gather and rot. Usually all components in the tank are of highly polished stainless steel to prevent the product from sticking.

Not to be forgotten is the corrosion resistance of the wetted parts. This is not simply a matter of their resistance to the products and cleaning agents. Under high temperatures, strong vibration and mechanical stress, electrochemical corrosion or intergranular corrosion may be enhanced. The one results in surface pitting, providing an ideal breeding place for bacteria, the other in the depletion of the nickel and chromium at the grain interfaces, which means the component will rust. Table 10.4 lists some stainless steels suitable for the food industry.

10.4.3 Process connections

The majority of process instruments are installed in pipes or tanks by means of threaded connections or flanges. Neither of these methods is suitable for food manufacture since both offer crevices and gaps where the product can accumulate and rot. In addition, the mounting and dismounting takes considerable effort, so cleaning becomes difficult.

Ideally, a process connection should offer no gaps where the product can become trapped. One solution is to weld the instrument in place and then grind and polish the inside of the connection. Unfortunately this means that the instrument cannot be exchanged should it fail. For thermowells, where the sensor insert is easily replaced, and for flowmeters, however, it is quite feasible and is often encountered.

Process instruments are generally installed by means of so-called sanitary couplings. These combine the need for a gap-free mounting with that of easy mounting and

Table 10.4 Stainless steels suitable for the food industry

Material	AISI*	Properties
1.4301	304	Good resistance against organic acids at moderate temperatures Good resistance against salt and alkalis at moderate temperatures
1.4404	316L	Increased resistance against non-oxidising acids such as acetic acid, tartaric acid, phosphoric acid Increased resistance against pitting and intercrystalline corrosion
1.4435	316L	Better corrosion resistance than Type 1.4404
1.4571	316 Ti	Increased corrosion resistance against particular acids and salt water Resistance against pitting corrosion

* The AISI steels are equivalents but do not have identical compositions

dismounting, allowing them to be quickly removed for cleaning. Over the years a number of different designs have come onto the market, a selection of which are summarised in Table 10.5.

10.4.4 Ingress and explosion protection

Just as important as the wetted parts and process connection is the design of the housing of a process instrument. Depending upon the instrument type, this may contain only the

Table 10.5 Sanitary couplings for the food industry

Type	Use	Description
Dairy coupling (to DIN 11851)	Pipes and tanks	Reasonably priced coupling that is frequently found in the food industry. Its weakness lies in the hygienic adaptation to the process which does not allow flush mounting. The coupling is made by a threaded boss and slotted sleeve. A conical seating and tapered nozzle with sealing ring comprise the process seal.
Aseptic coupling (to DIN 11864)	Pipes Tanks in preparation	Introduced in 1998 as a replacement for the dairy coupling. Designed to EHEDG recommendations, see Section 10.4.5, offering better hygiene thanks to a flush sealing construction. The mechanical coupling is via bolts or a threaded sleeve, the seal being flush with the pipe wall.
Varivent® coupling	Pipes	In-line housing that allows the flush mounting or the sensor, which is attached to the housing by means of a screw clamp. Three housing types cover a wide range of pipe diameters. For the majority of process sensors Type 3, for pipes of DN 40 upwards, is required. This facilitates the exchange of instruments.
APV coupling	Pipes	In-line housing of similar construction to the Varivent coupling. The sensor, however, is bolted in position.
SMS coupling	Pipes and tanks	Reasonably priced, Scandinavian standardised screw coupling which is also used in France. Its weakness lies in the hygienic adaptation to the process which does not allow flush mounting.
IDF coupling	Pipes and tanks	International Diary Federation screw coupling standardised in ISO
Tri-clamp® coupling	Pipes and tanks	Sanitary coupling with bevel seating produced by the Tri-Clover Company in America. Instruments are quickly mounted and fixed with snap-on clamps. The couplings find widespread use in America.

connecting terminals or the entire evaluating electronics. In both cases it must provide protection:

- from the ingress of dust or moisture from the outside
- when the sensor is used in an explosion hazardous area, from the egress of a spark or flame from the inside to the outside.

The former can be ensured by a suitable degree of protection, the latter by a suitable type of protection.

Ingress protection

As far as the ingress of dust and moisture is concerned, the world is divided into two camps. One half uses the IP standard (IEC 60 529) [8] and the other the American NEMA Standard No. 250. Nowadays, however, many manufacturers quote both in their technical specifications.

The IP standard is a description of the measures designed for the protection of the housing and the equipment within the housing. The degree of protection is indicated by a two-part code, e.g. IP 65. The first number is concerned with the protection from the ingress of solid matter, the second with water. As can be seen from Table 10.6, in order to withstand the frequent cleaning in a food production facility, housings with ratings of IP 65 or better are required.

The NEMA standard comprises 14 type codes which deal with practical requirements on housings suitable for indoor and outdoor use. It also makes a statement about the protection from external influences and conditions such as mechanical impact, corrosion, humidity, mould, pests, dust etc. As can be seen from Table 10.7, which lists only a selection of codes, a NEMA 4X enclosure is best suited to the requirements of the food industry.

Explosion protection

In comparison to the chemical industry, there is less need for explosion protection in the production of food. If flammable liquids or easily ignitable gases are present, however, then the instrumentation must be approved for use in explosion hazardous areas. Powders can also be a problem, since clouds of dust are easily combustible under certain conditions. For milling, storage, conveyance and bagging operations, therefore, the Dust-Ex equipment should be used. The types of explosion protection are standardised in EN 50 014 [9]. Table 10.8 lists and explains the ones of interest to food processing.

Table 10.6 Ingress protection categories to IEC 60 529

Code	Ingress protection against solids	Code	Ingress protection against water
0	Not protected	0	Not protected
1	$\geq$50 mm diameter, e.g. hand	1	Vertical dripping
2	$\geq$12.5 mm diameter, e.g. finger	2	Dripping (15° inclination)
3	$\geq$2.5 mm diameter, e.g. tool	3	Water spray
4	$\geq$1 mm diameter, e.g. wire	4	Splash water
5	Protected from dust	5	Jet of water
6	Dust-proof	6	Strong jet of water
		7	Temporary submersion
		8	Total submersion

Table 10.7 Degree of protection of enclosures as per NEMA Standard 250 (selection)

Type	Indoor	Outdoor	Degree of protection
1	yes		Protection against contact with equipment within the housing
2	yes		Protection against a specified quantity of water droplets and dirt
3		yes	Protection against blown dust, rain, sleet, rain and snow, and external ice formation
4	yes	yes	Protection against blown dust, jets splashes and jets of water
4X	yes	yes	Protection against corrosion, blown dust, jets splashes and jets of water
5	yes		Protection against dust falling, dirt and lubricating non-corrosive fluids
6	yes	yes	Protection against water penetration during occasional temporary submerging in limited depth

Table 10.8 Types of protection to EN 50 014

Type of protection	Significance
Intrinsic safety 'i'	Favoured by European manufacturers. The protection is achieved by limiting the current flowing in the device circuits. This means that there are restrictions regarding the installation and cabling.
Enhanced safety 'e'	The electronics are designed such that high temperatures and sparking cannot occur during normal operation. This type of protection is met, e.g. in 4-wire devices and power supplies.
Encapsulated 'm'	The entire electronics of the device are potted, so that no spark can enter the surounding atmosphere
Explosion-proof 'd'	The enclosure (housing) is designed such that any spark or explosion is retained within it. Frequently met in America where the entire cabling is routed in conduits. Also used in Europe for Dust-Ex protection

10.4.5 Standardisation authorities

As mentioned previously, the food industry is regulated by both government and industrial bodies. When choosing equipment, therefore, it pays to check whether a corresponding approval or authorisation has been granted. Table 10.9 lists several bodies which are of particular importance to safe food manufacture.

10.5 Applications

The production of food involves many processes. Each requires the measurement of particular process variables to ensure optimum operation conditions. The three examples here illustrate how the various components act together to produce the desired control.

10.5.1 Two-point temperature control

The majority of food manufacturing processes require at some point or other controlled heating or refrigeration. Two-point temperature control aims to keep the temperature of a

Table 10.9 Some regulatory bodies and associations important to process instrumentation

Body	Field of activity
3-A sanitary standards	Test and authorise process instrumentation regarding its fitness for use in food production, in particular the diary industry. The use of 3-A standards is entirely voluntary, but the quality is such that 3-A authorisation is accepted worldwide.
European Hygienic Equipment Design Group (EHEDG)	Issues guidelines on hygiene, tests and certifies equipment and publicises state-of-the-art technologies. An EHEDG approval means that the equipment at hand has been successfully tested with respect to its suitability for food applications
US Food and Drug Administration	Issues licences for products, whereby the processes, constituents, materials and constructional details are subject to examination. The use of an FDA-approved material, e.g. a stainless steel or elastamer, is a guarantee that the component concerned will not be attacked by the food product.
National Governmental Agencies	Lay down the permissible levels of trace organic and inorganic substances in food products. Since this is an area of permanent research, it is possible that regulations vary from country to country.
International Diary Federation	Responsible for standards and recommendations in the diary industry which are often published as ISO standards.

product within a specified range. The temperature sensor is built into the pipeline or tank where the product is to be found. The sensor measures the temperature and outputs either the temperature itself or an electrical signal from which it can be derived.

What happens next, depends upon the measuring system. In older analogue control systems, the signal is monitored by the controller. When the temperature drops below or exceeds the desired range, the controller acts to bring it back into range, e.g. by opening and closing valves. Modern transmitters and sensors with fieldbus output can often monitor up to four limits (HI_HI, HI, LO and LO_LO); violations are signalled by means of a bit array. The controller no longer has to monitor the limits, but reacts to a change of bit status. Alternatively the sensor itself takes over the control task and manipules the valves via a suitable control algorithm.

10.5.2 Pressure control in pipelines

Where food processing is based on liquid raw materials, e.g. milk or oil, or the final product is liquid, e.g. vegetable juices or yoghurt, then the transport through the plant must be carefully controlled. Milk, for instance, should be transported at a flowrate no greater than 2.5 m/s, otherwise the mechanical forces acting on it damage the product quality. As pressure drops in pipelines and across valves and manifolds are not constant, pressure control devices must be installed in the pipelines.

In general, pressure control is achieved by running the pumps at their operating speed and deciding on a pressure setpoint. The pressure sensor detects the actual pump pressure and passes the information on to the controller. This compares the actual value with the setpoint value and computes the setpoint deviation, which ideally is zero. If the deviation is not zero, the controller then sets the valves accordingly, so that the set pressure is maintained.

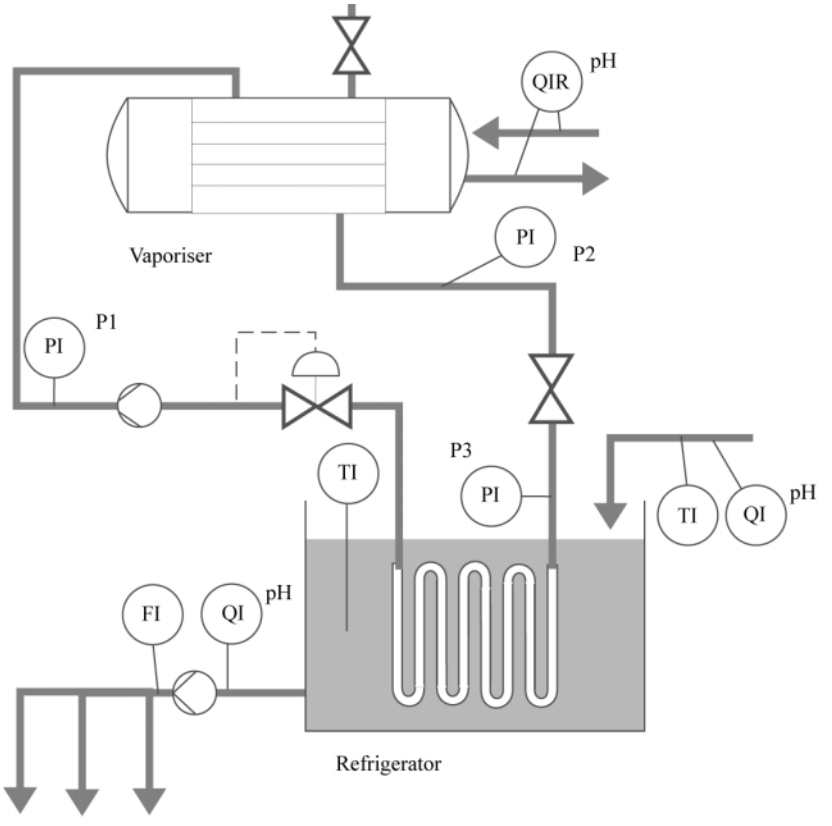


Fig. 10.8 Process instrumentation required for the control of an industrial refrigeration unit. FI = Flow indication, PI = Pressure indication, QI = Quality indication (pH), QIR = Quality indication and registration (pH), TI = Temperature indication.

10.5.3 Refrigeration plant

Refrigeration is required in many food processes, either during production or for the storage of the ingredients and end products. The standard ammonia gas refrigeration plant, see Fig. 10.8, comprises a compressor/condenser and vaporisor/heat exchanger together with valves to control the flow of ammonia around the refrigeration circuit.

A pump, operating at its rated speed, pumps ammonia into the compressor which comprises a bundle of pipes that are cooled by a continuous flow of water. The pump pressure $P1$ is monitored and controlled by the valve at the outlet side of the vaporiser. The cooled, pressurised ammonia liquefies in the condenser and flows through a second valve into the vaporiser. This comprises a cooling coil which is surrounded by the cooling medium, usually saltwater or a glycol/water mixture. As the ammonia enters the vaporiser it suddenly expands, changes from liquid to gas and as a result removes heat from its surroundings. The pressure of the liquid and gaseous ammonia $P2$ and $P3$ at both sides of the inlet valve is monitored, so that the pressure drop across the inlet valve can be optimised.

The cooling medium is pumped from the refrigerator to the various refrigeration coils in the plant. Its temperature is checked both at the outlet and the back flow inlet, the resultant information being used to control the refrigeration circuit. A pressure sensor

monitors the pumping circuit to ensure flow control as described in Section 10.5.2. The two pH sensors in the cooling circuit are used to monitor any ammonia leaks in the cooling coil by monitoring the difference between the pH at the inlet and outlet. A leak of a few parts per million causes a change of approximately 1 pH unit.

10.6 References

[1] *Food and Beverages, Measurement and Automation*, ed. M. Baur, Endress+Hauser, CH-4153 Reinach, Switzerland.

[2] IEC Standard 61 298, Parts 1–4, Process measurement and control devices, General methods and procedures for evaluating performance.
Part 1 (1995) General considerations.
Part 2 (1995) Tests under reference conditions.
Part 3 (1998) Tests for the effects of influence quantities.
Part 4 (1995) Evaluation report content.

[3] DIN 16 086, Electrical pressure measuring instruments; pressure sensors, pressure transmitters, pressure measuring instruments; concepts, specifications on data sheets, Beuth Verlag, Berlin, May 1992.

[4] IEC standard 60 751 (1983), Industrial platinum resistance thermometer sensors (currently under review).

[5] The International Temperature Scale of 1990 (ITS-90), see e.g. www.its-90.com.

[6] DIN 43 729, Electrical temperature sensor; connection heads for thermocouple-thermometers and resistance thermometers.

[7] IEC Standard 60 770, Transmitters for use in industrial-process control systems.
Part 1 (1999) Methods for performance evaluation.
Part 2 (1989) Guidance for inspection and routine testing (under review).

[8] IEC Standard 60 529, Degrees of protection provided by enclosures (IP code).
Issued 1989, Amendment (1999)

[9] EN 50 014, Electrical apparatus for potentially explosive atmospheres – General requirements.